

Information–Theoretic Barriers for Collatz Proofs

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Machine-Checked Formalization in Isabelle/HOL

Abstract

In an earlier paper, *The Collatz $3n + 1$ Conjecture is Unprovable* (2012), the author argued that any proof of the Collatz conjecture would need to contain arbitrarily large amounts of information about the parity pattern of Collatz trajectories, making a finite proof impossible. The present work formalises that idea in Isabelle/HOL. Under explicit assumptions about how proofs store and preserve information, we prove that no finite proof can establish the required convergence property. This yields a machine-checked information-theoretic barrier theorem inspired by the earlier argument.

Provenance

The present development formalises the information-theoretic ideas underlying the author’s earlier paper:

C. A. Feinstein, “*The Collatz $3n + 1$ Conjecture is Unprovable*”, arXiv:math/0312309; *Global Journal of Science Frontier Research*, Mathematics and Decision Sciences, Volume 12, Issue 8 (2012), 13–15.

The assumptions required for that argument are made explicit and formalised within Isabelle/HOL, yielding a machine-checked theorem showing that no finite proof can exist within the corresponding class of proof systems. These assumptions are motivated by structural properties of the Collatz map.

The author of this formalisation received assistance from two AI systems — ChatGPT (OpenAI) and Claude (Anthropic). Their assistance consisted of drafting and refining explanatory text, improving the readability of the introduction and comments, and helping diagnose or structure Isabelle/HOL proof scripts.

Main goal

The present development establishes an information-theoretic barrier for proof methods that explicitly store the required parity information.

High-level strategy

The argument formalised here isolates the following phenomena:

1. If a proposed proof has length L , an incompressible parity vector of length $L + 1$ can be chosen.
2. The Collatz map realises that vector as the first $L + 1$ parities of a positive trajectory, while also making the parities at steps L and $L + 1$ equal.
3. The equality of these two parities implies that the value at step L is greater than 2, and therefore any step at which the trajectory equals 1 must occur after step L .
4. Under the trace-specification assumption, the proposed proof must contain an encoding of the chosen $L + 1$ bits, contradicting its length L .

The structure of the argument is inspired by Chaitin-style incompressibility methods, but is applied to the representation of computational traces within proof certificates rather than to the computation of specific strings.

Structure of the formalisation

1. **Collatz map and parity vectors** We define the Collatz function T and formalise parity vectors as computational traces of the iterative process.
2. **Affine formula characterisation** We show that $T^{(k)}(n)$ admits an affine representation $(3^s \cdot n + c)/2^k$, and prove that the parameters (k, s, c) are uniquely determined by the parity vector.
3. **Two-adic invariance** We establish a key invariance property: adding 2^k to a starting value does not affect the first k parity bits. This property underlies the realisability construction.
4. **Every parity vector is realisable** We prove that every finite binary string occurs as the parity vector of some starting value. This shows that arbitrarily complex computational traces are inherent to the Collatz dynamics.
5. **Proof system setup** We introduce an abstract model of proof certificates based on bitstrings, substring containment, and information-theoretic compressibility.
6. **The original information-barrier argument** We formalise the original information-theoretic argument for the Collatz conjecture, with its proof-system requirements stated explicitly.

```
theory Collatz_Information_Barriers
  imports Main
begin
```

1 Collatz map and parity vectors

The Collatz function

We define the Collatz function T by

$$T(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ (3n + 1)/2 & \text{if } n \text{ is odd.} \end{cases}$$

```
definition T :: "nat ⇒ nat" where
  "T n = (if even n then n div 2 else (3*n + 1) div 2)"
```

```
abbreviation Tpow :: "nat ⇒ nat ⇒ nat"
  where "Tpow k n ≡ (T ^^ k) n"
```

The Collatz conjecture

The formulation of the conjecture used in the earlier paper, and throughout this development, is:

$$\forall n > 0. \exists k. T^{(k)}(n) = 1.$$

The parity vector as computational trace

The parity vector *parity_vec n k* records the sequence of even and odd values encountered during the first *k* iterations starting from *n*:

$$\text{parity_vec } n \ k = [\text{odd}(n), \text{odd}(T(n)), \text{odd}(T^{(2)}(n)), \dots, \text{odd}(T^{(k-1)}(n))].$$

This parity vector represents the *computational trace*: the sequence of branching decisions made while repeatedly applying the Collatz map.

Example

For $n = 27$ we have:

$$T(27) = 41, \quad T^{(2)}(27) = 62, \quad T^{(3)}(27) = 31.$$

Thus:

$$\begin{aligned} \text{parity_vec } 27 \ 4 &= [\text{odd}(T^{(0)}(27)), \text{odd}(T^{(1)}(27)), \text{odd}(T^{(2)}(27)), \text{odd}(T^{(3)}(27))] \\ &= [\text{odd}(27), \text{odd}(41), \text{odd}(62), \text{odd}(31)] \\ &= [\text{True}, \text{True}, \text{False}, \text{True}]. \end{aligned}$$

definition collatz_conjecture :: bool where
`"collatz_conjecture <=> (forall n>0. exists k. Tpow k n = 1)"`

definition parity_vec :: "nat => nat => bool list"
`where "parity_vec n k = map (lambda i. odd (Tpow i n)) [0..<k]"`

lemma length_parity_vec[simp]: "length (parity_vec n k) = k"
`by (simp add: parity_vec_def)`

2 Affine formula characterisation

After k iterations of the Collatz map T , the result admits an affine representation:

$$T^{(k)}(n) = \frac{3^s \cdot n + c}{2^k}.$$

Here:

- k is the number of iterations.
- s counts the number of odd values in the parity vector, that is, the number of steps of the form $(3n + 1)/2$.
- c is a constant determined by the specific parity sequence.

Intuition

Each odd step multiplies the current value by 3, adds 1, and then divides by 2, while each even step simply divides by 2. After k steps, the cumulative effect is multiplication by $3^s/2^k$ and the addition of $c/2^k$.

Uniqueness

The parameters (k, s, c) are uniquely determined by the parity vector. In particular, the parity vector encodes all computational information needed to reconstruct the affine formula for $T^{(k)}(n)$.

```
fun params0 :: "bool list ⇒ nat × nat" where
  "params0 [] = (0, 0)" |
  "params0 (b # bs) =
    (let (c,s) = params0 bs in
     if b then (2*c + 3^s, Suc s) else (2*c, s))"

definition formula_of :: "bool list ⇒ nat × nat × nat" where
  "formula_of x = (length x, snd (params0 x), fst (params0 x))"

lemma params0_odd_count:
  "snd (params0 xs) = length (filter id xs)"
proof (induction xs)
  case Nil
  show ?case by simp
next
  case (Cons b bs)
  obtain c s where P: "params0 bs = (c,s)"
    by (cases "params0 bs") auto
  have S: "s = length (filter id bs)"
    using Cons.IH by (simp add: P id_def)
  show ?case by (cases b) (simp_all add: P S id_def)
qed

lemma params0_two_odd_steps:
  "params0 [True, True] = (5, 2)"
  by simp
```

Injectivity

Different parity vectors produce distinct triples (k, s, c) in the representation

$$T^{(k)}(n) = \frac{3^s \cdot n + c}{2^k}.$$

```
lemma params0_injective_len:
  assumes "length xs = length ys" and "params0 xs = params0 ys"
  shows "xs = ys"
```

```

    using assms
  proof (induction xs arbitrary: ys)
    case Nil
    then show ?case by (cases ys) auto
  next
    case (Cons a xs)
    from Cons.prem1 obtain b zs where Y: "ys = b # zs"
      by (cases ys) auto
    obtain c s where X: "params0 xs = (c,s)"
      by (cases "params0 xs") auto
    obtain d t where Z: "params0 zs = (d,t)"
      by (cases "params0 zs") auto
    have E:
      "(if a then (2*c + 3^s, Suc s) else (2*c,s)) =
       (if b then (2*d + 3^t, Suc t) else (2*d,t))"
      using Cons.prem2 by (simp add: Y X Z)
    have H: "a = b"
    proof -
      have "odd (fst (if a then (2*c + 3^s, Suc s) else (2*c,s))) =
        odd (fst (if b then (2*d + 3^t, Suc t) else (2*d,t)))"
        using E by simp
      then show ?thesis by (cases a; cases b) simp_all
    qed
    have ST: "s = t"
      using E H by (cases a; cases b) auto
    have CD: "c = d"
      using E H ST by (cases a; cases b) auto
    have P: "params0 xs = params0 zs"
      using X Z CD ST by simp
    have L: "length xs = length zs"
      using Cons.prem1 by (simp add: Y)
    have "xs = zs" using Cons.IH[OF L P] .
    then show ?case using H Y by simp
  qed

lemma formula_determines_parity_on_len:
  assumes "length x = k" "length y = k"
  and "formula_of x = formula_of y"
  shows "x = y"
proof -
  have P: "params0 x = params0 y"
    using assms(3)
    by (cases "params0 x"; cases "params0 y";
        simp add: formula_of_def)
  have L: "length x = length y"
    using assms(1,2) by simp
  show ?thesis
    by (rule params0_injective_len[OF L P])
qed

```

3 Two-adic invariance

Key lemma

Adding $q \cdot 2^k$ to m preserves the first k parity bits:

$$\text{parity_vec}(m + q \cdot 2^k, k) = \text{parity_vec}(m, k).$$

Intuition

The offset $q \cdot 2^k$ is beyond the resolution of the first k steps. After one application of the Collatz map T , the offset becomes $q \cdot 2^{k-1}$ if m is even, or $3q \cdot 2^{k-1}$ if m is odd, and therefore remains a multiple of 2^{k-1} . This behaviour continues inductively through subsequent iterations.

Consequence

This invariance property enables realisability. One can tune a number to have any desired parity sequence by adding suitable multiples of 2^k , without affecting parity bits that have already been fixed.

```
lemma T_funpow_commute: "T ((T ^^ j) n) = (T ^^ j) (T n)"
proof (induction j)
  case 0
  show ?case by simp
next
  case (Suc j)
  have "T ((T ^^ Suc j) n) = T (T ((T ^^ j) n))" by simp
  also have "... = T ((T ^^ j) (T n))" by (simp add: Suc.IH)
  also have "... = (T ^^ Suc j) (T n)" by simp
  finally show ?case .
qed

lemma parity_vec_Suc:
  "parity_vec n (Suc k) = odd n # parity_vec (T n) k"
proof (rule nth_equalityI)
  show "length (parity_vec n (Suc k)) =
        length (odd n # parity_vec (T n) k)"
    by simp
next
  fix i assume iLt: "i < length (parity_vec n (Suc k))"
  then have iSk: "i < Suc k" by (simp add: parity_vec_def)
  consider (Z) "i = 0" | (S) j where "i = Suc j" "j < k"
    using iSk by (cases i) auto
  then show "parity_vec n (Suc k) ! i =
            (odd n # parity_vec (T n) k) ! i"
    proof cases
      case Z
      show ?thesis
```



```

    using Z by (simp add: parity_vec_def del: upt_Suc)
next
  case (S j)
  show ?thesis
    using S
    by (simp add: parity_vec_def T_funpow_commute del: upt_Suc)
qed
qed

lemma T_even_identity:
  assumes "even n"
  shows "2 * T n = n"
proof -
  obtain q where N: "n = 2*q" using assms by (elim evenE)
  show ?thesis by (simp add: N T_def)
qed

lemma T_odd_identity:
  assumes "odd n"
  shows "2 * T n = 3*n + 1"
proof -
  obtain q where N: "n = 2*q + 1" using assms by (elim oddE)
  show ?thesis by (simp add: N T_def algebra_simps)
qed

lemma collatz_affine_identity:
  "2^k * Tpow k n =
   3^(snd (params0 (parity_vec n k))) * n +
   fst (params0 (parity_vec n k))"
proof (induction k arbitrary: n)
  case 0
  show ?case by (simp add: parity_vec_def)
next
  case (Suc k)
  obtain c s where P: "params0 (parity_vec (T n) k) = (c,s)"
    by (cases "params0 (parity_vec (T n) k)") auto
  have IH: "2^k * Tpow k (T n) = 3^s * T n + c"
    using Suc.IH[of "T n"] by (simp add: P)
  have D:
    "2^(Suc k) * Tpow (Suc k) n = 3^s * (2 * T n) + 2*c"
  proof -
    have "2^(Suc k) * Tpow (Suc k) n =
      2 * (2^k * Tpow k (T n))"
      by (simp add: T_funpow_commute algebra_simps)
    also have "... = 2 * (3^s * T n + c)" by (simp only: IH)
    also have "... = 3^s * (2 * T n) + 2*c" by (simp add: algebra_simps)
    finally show ?thesis .
  qed
  show ?case

```

```

proof (cases "even n")
  case True
    have Q: "params0 (parity_vec n (Suc k)) = (2*c,s)"
      by (simp add: parity_vec_Suc P True)
    show ?thesis using D T_even_identity[OF True] by (simp add: Q)
  next
    case False
    have O: "odd n" using False by simp
    have Q: "params0 (parity_vec n (Suc k)) =
      (2*c + 3^s, Suc s)"
      by (simp add: parity_vec_Suc P False)

    have R:
      "2^(Suc k) * Tpow (Suc k) n =
      3^s * (3*n + 1) + 2*c"
      using D
      by (simp only: T_odd_identity[OF O])

    show ?thesis
      using R by (simp add: Q algebra_simps)
  qed
qed

lemma parity_vec_add_pow2_invariant:
  fixes m q k :: nat
  shows "parity_vec (m + q * 2 ^ k) k = parity_vec m k"
proof (induction k arbitrary: m q)
  case 0
    show ?case by (simp add: parity_vec_def)
  next
    case (Suc k)
    let ?Δ = "q * 2 ^ Suc k"

    have head: "odd (m + ?Δ) = odd m"
      by simp

    have tail: "parity_vec (T (m + ?Δ)) k = parity_vec (T m) k"
    proof (cases "even m")
      case True
        then obtain t where m2: "m = 2*t" by (elim evenE)
        have "(m + ?Δ) div 2 = t + q * 2 ^ k"
          by (simp add: m2)
        hence "T (m + ?Δ) = t + q * 2 ^ k"
          using True by (simp add: T_def)
        moreover have "T m = t"
          using True m2 by (simp add: T_def)
        ultimately show ?thesis
          using Suc.IH[of t q] by simp
      case False
    next

```

```

case False
then obtain t where m2: "m = 2*t + 1" by (elim oddE)
have "(3*(m + ?Δ) + 1) div 2
      = (3*m + 1) div 2 + (3*q) * 2 ^ k"
proof -
  have "3*(m + ?Δ) + 1 = (3*m + 1) + (3*q) * 2 ^ Suc k"
    by simp
  moreover have "even (3*m + 1)"
    using False m2 by simp
  ultimately show ?thesis by simp
qed
hence "T (m + ?Δ) = T m + (3*q) * 2 ^ k"
  using False by (simp add: T_def)
thus ?thesis
  using Suc.IH[of "T m" "3*q"] by simp
qed

have step1: "parity_vec (m + ?Δ) (Suc k) =
             odd (m + ?Δ) # parity_vec (T (m + ?Δ)) k"
  by (simp add: parity_vec_Suc)
also have step2: "... = odd m # parity_vec (T m) k"
  using head tail by blast
also have step3: "... = parity_vec m (Suc k)"
  by (simp add: parity_vec_Suc)
finally show ?case .
qed

```

4 Every parity vector is realisable

Realisability

For any finite binary string x , there exists a natural number n whose parity sequence agrees with x :

$$\forall x. \exists n. \text{parity_vec } n \text{ (length } x) = x.$$

This means that every possible computational trace of the Collatz map actually occurs for some starting number. In particular, parity traces of arbitrary length, including incompressible traces, are realised.

Helper lemmas for modular arithmetic

The following lemmas establish basic modular-arithmetic facts that are used in the realisability construction.

```

lemma power_mod_nat:
  fixes a m n :: nat
  shows "(a ^ n) mod m = ((a mod m) ^ n) mod m"

```

```

proof (induction n)
  case 0
  show ?case by simp
next
  case (Suc n)
  have "(a ^ Suc n) mod m = (a * a ^ n) mod m" by simp
  also have "... = (((a mod m) * a ^ n) mod m)"
    by (simp add: mod_mult_left_eq)
  also have "... = (((a mod m) * ((a ^ n) mod m)) mod m)"
    by (simp add: mod_mult_right_eq)
  also have "... = (((a mod m) * (((a mod m) ^ n) mod m)) mod m)"
    by (simp add: Suc.IH)
  also have "... = (((a mod m) * ((a mod m) ^ n)) mod m)"
    by (simp add: mod_mult_right_eq)
  also have "... = ((a mod m) ^ Suc n) mod m" by simp
  finally show ?case .
qed

```

```

lemma power_mult_nat:
  fixes a :: nat
  shows "a ^ (m * n) = (a ^ m) ^ n"
proof (induction n)
  case 0
  show ?case by simp
next
  case (Suc n)
  have "a ^ (m * Suc n) = a ^ (m*n + m)" by (simp add: add.commute)
  also have "... = a ^ (m*n) * a ^ m" by (simp add: power_add)
  also have "... = (a ^ m) ^ n * a ^ m" by (simp add: Suc.IH)
  also have "... = (a ^ m) ^ Suc n" by simp
  finally show ?case .
qed

```

```

lemma pow4_mod3: "((4::nat) ^ m) mod 3 = 1"
  by (simp add: power_mod_nat)

```

```

lemma pow2_mod3_even:
  assumes "even 1"
  shows "(2 :: nat) ^ 1 mod 3 = 1"
proof -
  obtain m where L: "1 = 2*m" using assms by (erule evenE)
  have "2 ^ 1 mod 3 = (2 ^ (2*m)) mod 3" by (simp add: L)
  also have "... = (((2::nat) ^ 2) ^ m) mod 3" by (simp add: power_mult_nat)
  also have "... = (4 ^ m) mod 3" by simp
  also have "... = 1" by (rule pow4_mod3)
  finally show ?thesis .
qed

```

```

lemma pow2_mod3_odd:
  assumes "odd 1"
  shows "(2 :: nat) ^ 1 mod 3 = 2"
proof -
  obtain m where L: "1 = Suc (2*m)" using assms
  by (metis Suc_eq_plus1 oddE)
  have "2 ^ 1 mod 3 = (2 * 2 ^ (2*m)) mod 3" by (simp add: L)
  also have "... = (2 * ((2::nat) ^ 2) ^ m) mod 3"
  by (simp add: power_mult_nat)
  also have "... = (2 * ((4 ^ m) mod 3)) mod 3"
  by (simp add: mod_mult_right_eq)
  also have "... = (2 * 1) mod 3" by (simp add: pow4_mod3)
  also have "... = 2" by simp
  finally show ?thesis .
qed

lemma pow2_mod3:
  fixes 1 :: nat
  shows "(2 :: nat) ^ 1 mod 3 = (if even 1 then 1 else 2)"
  by (cases "even 1") (simp add: pow2_mod3_even, simp add: pow2_mod3_odd)

lemma choose_t_even_mod3:
  fixes m0 1 :: nat
  assumes "even 1"
  shows "∃ t ≤ 2. (m0 + t * (2 :: nat) ^ 1) mod 3 = 2"
proof -
  define t where "t = (2 - (m0 mod 3)) mod 3"
  have t_le2: "t ≤ 2" by (simp add: t_def)
  have "(m0 + t * 2 ^ 1) mod 3
    = (m0 mod 3 + t * (2 ^ 1 mod 3)) mod 3"
  by (metis mod_add_cong mod_mod_trivial mod_mult_right_eq)
  also have "... = (m0 mod 3 + t) mod 3"
  using assms by (simp add: pow2_mod3)
  also have "... = (m0 mod 3 + ((2 - (m0 mod 3)) mod 3)) mod 3"
  by (simp add: t_def)
  also have "... = 2"
  by (cases "m0 mod 3") simp_all
  finally have targ: "(m0 + t * 2 ^ 1) mod 3 = 2" .
  from t_le2 targ show ?thesis by blast
qed

lemma choose_t_odd_mod3:
  fixes m0 1 :: nat
  assumes "odd 1"
  shows "∃ t ≤ 2. (m0 + t * (2 :: nat) ^ 1) mod 3 = 2"
proof -
  define t where "t = (2 * (2 - (m0 mod 3))) mod 3"
  have t_le2: "t ≤ 2" by (simp add: t_def)
  have "(m0 + t * 2 ^ 1) mod 3

```

```

      = (m0 mod 3 + t * (2 ^ 1 mod 3)) mod 3"
    by (metis mod_add_cong mod_mod_trivial mod_mult_right_eq)
  also have "... = (m0 mod 3 + ((2 * (2 - (m0 mod 3))) mod 3) * 2)
    mod 3"
    by (simp add: t_def pow2_mod3 assms)
  also have "... = (m0 mod 3 + (4 * (2 - (m0 mod 3))) mod 3) mod 3"
    using mod_mult_right_eq by (metis (no_types, lifting)
      distrib_right mod_add_eq mod_add_left_eq mult_2_right numeral_Bit0_eq_double)
  also have "... = (m0 mod 3 + ((4 mod 3) *
    ((2 - (m0 mod 3)) mod 3)) mod 3) mod 3"
    using mod_mult_left_eq by (metis mod_mult_right_eq)
  also have "... = (m0 mod 3 + ((2 - (m0 mod 3)) mod 3)) mod 3" by simp
  also have "... = 2" by (cases "m0 mod 3") simp_all
  finally have targ: "(m0 + t * 2 ^ 1) mod 3 = 2" .
  show ?thesis using t_le2 targ by blast
qed

lemma parity_vector_realizable:
  fixes x :: "bool list"
  shows "∃n. parity_vec n (length x) = x"
proof (induction x)
  case Nil
  show ?case by (metis length_0_conv length_parity_vec)
next
  case (Cons b bs)
  obtain m0 where IH: "parity_vec m0 (length bs) = bs"
    using Cons.IH by blast
  show ?case
  proof (cases b)
    case False
    let ?n = "2*m0"
    have "parity_vec ?n (length (b # bs))
      = odd ?n # parity_vec (T ?n) (length bs)"
      by (simp add: parity_vec_Suc)
    also have "... = False # bs" by (simp add: T_def IH)
    also have "... = b # bs" by (simp add: False)
    finally show ?thesis by (intro exI[of _ ?n])
  next
    case True
    let ?l = "length bs"
    obtain t where t_le2: "t ≤ 2" and
      targ: "(m0 + t * 2 ^ ?l) mod 3 = 2"
      by (cases "even ?l")
      (use choose_t_even_mod3[of ?l m0]
        choose_t_odd_mod3[of ?l m0] in auto)
    let ?m = "m0 + t * 2 ^ ?l"
    have tail_preserved: "parity_vec ?m ?l = bs"
      using IH parity_vec_add_pow2_invariant[of m0 t ?l] by simp
    define q where "q = ?m div 3"

```

```

have m_eq: "?m = 3*q + 2"
proof -
  have "?m = 3 * (?m div 3) + (?m mod 3)" by (simp add: div_mult_mod_eq)
  also have "... = 3*q + 2" by (simp add: q_def targ)
  finally show ?thesis .
qed
let ?n = "(2 * ?m - 1) div 3"
have head: "odd ?n"
proof -
  have "?n = (2 * (3*q + 2) - 1) div 3" by (simp add: m_eq)
  also have "... = (6*q + 3) div 3" by simp
  also have "... = 2*q + 1" by simp
  finally show ?thesis by simp
qed
have tail_step: "parity_vec (T ?n) ?l = bs"
  using tail_preserved head by (simp add: m_eq T_def)
have "parity_vec ?n (length (b # bs)) =
  odd ?n # parity_vec (T ?n) ?l"
  by (simp add: parity_vec_Suc)
also have "... = True # bs" using head tail_step by simp
also have "... = b # bs" by (simp add: True)
finally show ?thesis by (intro exI[of _ ?n])
qed
qed

```

Realisability with matching successive parity

Let x be a bit vector of length $L + 1$. There is a positive natural number n such that

$$x = (n, T(n), \dots, T^{(L)}(n)) \pmod{2}$$

and

$$T^{(L+1)}(n) = T^{(L)}(n) \pmod{2}.$$

To obtain matching parities at steps L and $L + 1$, append a copy of the final bit of x to the vector. Parity-vector realisability, together with two-adic invariance, then supplies a positive starting value having this extended parity vector.

Since n is positive, every value in its trajectory is positive. If $T^{(L)}(n) = 1$, then $T^{(L+1)}(n) = 2$; if $T^{(L)}(n) = 2$, then $T^{(L+1)}(n) = 1$. In either case, the two values have opposite parities. Their equal parities therefore imply that $T^{(L)}(n) > 2$. If the trajectory had reached 1 at some step $k \leq L$, every subsequent value through step L would belong to the cycle $1, 2, 1, 2, \dots$, which would give $T^{(L)}(n) \leq 2$. This contradicts $T^{(L)}(n) > 2$. Therefore, $T^{(k)}(n) = 1$ implies $k > L$.

lemma `parity_vector_realizable_with_matching_next_parity:`
`assumes x_len: "length x = Suc L"`

```

shows "∃n>0.
      parity_vec n (Suc L) = x ∧
      odd (Tpow (Suc L) n) = odd (Tpow L n)"
proof -
  let ?y = "x @ [x ! L]"
  obtain m where m_realizes:
    "parity_vec m (length ?y) = ?y"
    using parity_vector_realizable[of ?y] by blast
  define n where "n = m + 2 ^ Suc (Suc L)"
  have y_len: "length ?y = Suc (Suc L)"
    using x_len by simp
  have m_realizes':
    "parity_vec m (Suc (Suc L)) = ?y"
    using m_realizes y_len by simp
  have invariant:
    "parity_vec n (Suc (Suc L)) =
     parity_vec m (Suc (Suc L))"
    using parity_vec_add_pow2_invariant[of m 1 "Suc (Suc L)"]
    by (simp add: n_def)
  have n_realizes:
    "parity_vec n (Suc (Suc L)) = ?y"
    using invariant m_realizes' by simp
  have n_pos: "n > 0"
    by (simp add: n_def)
  have prefix:
    "parity_vec n (Suc L) = x"
  proof -
    have "parity_vec n (Suc L) =
          take (Suc L) (parity_vec n (Suc (Suc L)))"
      by (simp add: parity_vec_def take_map)
    also have "... = take (Suc L) ?y"
      by (simp add: n_realizes)
    also have "... = x"
      using x_len by simp
    finally show ?thesis .
  qed
  have at_L:
    "odd (Tpow L n) = x ! L"
  proof -
    have x_nonempty: "x ≠ []"
      using x_len by auto

    have "odd (Tpow L n) =
          last (parity_vec n (Suc L))"
      by (simp add: parity_vec_def)
    also have "... = last x"
      by (simp add: prefix)
    also have "... = x ! (length x - 1)"
      by (rule last_conv_nth[OF x_nonempty])
  qed

```



```

    also have "... = x ! L"
    using x_len by simp
    finally show ?thesis .
qed

have at_Suc_L:
  "odd (Tpow (Suc L) n) = x ! L"
proof -
  have "odd (Tpow (Suc L) n) =
    last (parity_vec n (Suc (Suc L)))"
    by (simp add: parity_vec_def)
  also have "... = last ?y"
    by (simp add: n_realizes)
  also have "... = x ! L"
    by simp
  finally show ?thesis .
qed
show ?thesis
  using n_pos prefix at_L at_Suc_L by blast
qed

lemma T_positive:
  assumes "n > 0"
  shows "T n > 0"
proof (cases "even n")
  case True
  then obtain q where n_eq: "n = 2 * q"
    by (elim evenE)
  have "q > 0"
    using assms n_eq by simp
  then show ?thesis
    using True by (simp add: T_def n_eq)
next
  case False
  then obtain q where n_eq: "n = 2 * q + 1"
    by (elim oddE)
  then show ?thesis
    by (simp add: T_def)
qed

lemma Tpow_positive:
  assumes "n > 0"
  shows "Tpow k n > 0"
  using assms
proof (induction k arbitrary: n)
  case 0
  then show ?case by simp
next
  case (Suc k)

```

```

have iter_pos: "Tpow k n > 0"
  using Suc.IH Suc.prem by blast
have "T (Tpow k n) > 0"
  using T_positive[OF iter_pos] .
then show ?case
  by (simp add: funpow_Suc_right)
qed

lemma equal_successive_parity_imp_gt_two:
  assumes n_pos: "n > 0"
  and same_parity:
    "odd (Tpow (Suc L) n) = odd (Tpow L n)"
  shows "Tpow L n > 2"
proof -
  have step:
    "Tpow (Suc L) n = T (Tpow L n)"
  by (simp add: funpow_Suc_right)
  have iter_pos: "Tpow L n > 0"
  using Tpow_positive n_pos by blast
  have not_one: "Tpow L n  $\neq$  1"
  using same_parity step by (auto simp: T_def)
  have not_two: "Tpow L n  $\neq$  2"
  using same_parity step by (auto simp: T_def)
  show ?thesis
  using iter_pos not_one not_two by linarith
qed

lemma Tpow_one_le_two:
  "Tpow j 1  $\leq$  2"
proof -
  have "Tpow j 1 = 1  $\vee$  Tpow j 1 = 2"
  proof (induction j)
    case 0
    then show ?case by simp
  next
    case (Suc j)
    then show ?case
      by (auto simp: funpow_Suc_right T_def)
  qed
  then show ?thesis by auto
qed

lemma reaches_one_only_after_L:
  assumes n_pos: "n > 0"
  and same_parity:
    "odd (Tpow (Suc L) n) = odd (Tpow L n)"
  and reaches: "Tpow k n = 1"
  shows "k > L"
proof (rule ccontr)

```

```

assume "¬ k > L"
then have k_le: "k ≤ L" by simp
have L_decomp: "L = (L - k) + k"
  using k_le by simp
have "Tpow L n = Tpow ((L - k) + k) n"
  by (rule arg_cong[OF L_decomp])
also have "... = Tpow (L - k) (Tpow k n)"
  by (simp add: funpow_add)
also have "... = Tpow (L - k) 1"
  by (simp add: reaches)
also have "... ≤ 2"
  by (rule Tpow_one_le_two)
finally have "Tpow L n ≤ 2" .
moreover have "Tpow L n > 2"
  using equal_successive_parity_imp_gt_two[OF n_pos same_parity] .
ultimately show False by simp
qed

```

5 Proof system setup

```

type_synonym bit = bool
type_synonym bitstring = "bit list"

```

Substring containment

We introduce a substring-containment relation as a *concrete and explicit* model of proofs that store computational trace information as literal data. Formally, a bitstring p contains a bitstring s if s occurs verbatim as a contiguous substring of p , i.e. if there exist bitstrings u and v such that $p = u@s@v$.

Remark. The choice of substring containment is deliberately strong. It provides a simple, syntactic notion of explicit information storage that is easy to reason about formally and avoids ambiguity about how information is represented inside a proof certificate. In the main barrier theorem, literal containment implies that the encoded parity vector cannot be longer than the proof certificate. The separately established incompressibility condition ensures that, under the chosen encoding, the encoded vector is at least as long as the vector itself.

```

definition contains :: "bitstring ⇒ bitstring ⇒ bool"
  where "contains p s ⟷ (∃ u v. p = u @ s @ v)"

```

```

lemma contains_len_bound: "contains p s ==> length s <= length p"
  by (auto simp: contains_def)

```

6 The unprovability theorem

```

lemma finite_bitstrings_of_len:
  "finite {s::bitstring. length s = m}"
proof -
  have fin_aux:
    "finite {s::bool list. set s <= (UNIV::bool set) ∧ length s = m}"
    by (rule finite_lists_length_eq) simp
  have "{s::bitstring. length s = m}
    = {s. set s <= (UNIV::bool set) ∧ length s = m}"
    by auto
  then show ?thesis using fin_aux by (simp only:)
qed

lemma many_strings_of_length:
  "card {s::bitstring. length s = m} = 2 ^ m"
proof (induction m)
  case 0
  have "{s::bitstring. length s = 0} = {[]}" by auto
  thus ?case by simp
next
  case (Suc m)
  have "{s. length s = Suc m} =
    (%b. True # b) ` {s. length s = m}
    Un (%b. False # b) ` {s. length s = m}"
    by (auto simp: length_Suc_conv)
  moreover have "(\b. True # b) ` {s. length s = m}
    Int (\b. False # b) ` {s. length s = m} = {}"
    by auto
  moreover have "inj_on (\b. True # b) {s. length s = m}"
    by (auto simp: inj_on_def)
  moreover have "inj_on (\b. False # b) {s. length s = m}"
    by (auto simp: inj_on_def)
  ultimately show ?case
    using Suc.IH card_Un_disjoint card_image
    by (smt (verit) Suc_1 Suc_pred card.infinite diff_add_zero
      mult_2 nat.discI plus_1_eq_Suc power_Suc0_right power_add
      power_eq_0_iff zero_less_one)
qed

definition compressible :: "bitstring ⇒ (bitstring ⇒ bitstring) ⇒ bool"
where
  "compressible s enc ≡ length (enc s) < length s"

definition incompressible_by :: "bitstring ⇒ (bitstring ⇒ bitstring)
⇒ bool" where
  "incompressible_by s enc ≡ ¬compressible s enc"

lemma sum_pow2_lt: "sum (%i. (2::nat) ^ i) {..

```

```

    by (induction m) simp_all

lemma finite_bitstrings_le_len:
  "finite {s::bitstring. length s ≤ m}"
proof (induction m)
  case 0 show ?case by (simp add: finite_bitstrings_of_len)
next
  case (Suc m)
  have "{s::bitstring. length s ≤ Suc m}
    = {s. length s ≤ m} Un {s. length s = Suc m}" by auto
  thus ?case using Suc.IH finite_bitstrings_of_len by (simp add: finite_UnI)
qed

lemma card_bitstrings_le_len:
  "card {s::bitstring. length s ≤ m} = sum (%i. 2 ^ i) {..m}"
proof -
  let ?S = "λi. {s::bitstring. length s = i}"
  have union_eq: "{s::bitstring. length s ≤ m} =
    Union ((%i. ?S i) ` {..m})"
    by auto
  have fin_index: "finite ({..m}::nat set)" by simp
  have fin_each: "!!i. i ≤ m ==> finite (?S i)"
    by (simp add: finite_bitstrings_of_len)
  have disj: "!!i j. i ≤ m ==> j ≤ m ==> i ~ j
    ==> ?S i Int ?S j = {}"
    by auto
  have "card (Union ((%i. ?S i) ` {..m})) = sum (%i. card (?S i)) {..m}"
    by (rule card_UN_disjoint) (use fin_index fin_each disj in auto)
  also have "... = sum (%i. 2 ^ i) {..m}"
    by (simp add: many_strings_of_length)
  finally show ?thesis
    by (simp add: union_eq)
qed

```

The pigeonhole argument

Theorem. For any injective encoding, there exists an incompressible bit-string.

```

theorem incompressible_strings_exist_for_enc:
  fixes enc :: "bitstring ⇒ bitstring"
  assumes "inj enc"
  shows "∃ s. length s = m ∧ incompressible_by s enc"
proof (cases m)
  case 0
  have "length ([:bitstring]) = 0 ∧ 0 ≤ length (enc [])"
    by simp
  then show ?thesis using 0 incompressible_by_def compressible_def by
  auto

```

```

next
  case (Suc r)
  let ?S = "{t::bitstring. length t = Suc r}"
  let ?T = "{u::bitstring. length u ≤ r}"
  have finS: "finite ?S" by (simp add: finite_bitstrings_of_len)
  have finT: "finite ?T" by (simp add: finite_bitstrings_le_len)
  have Sm: "card ?S = 2 ^ Suc r"
    by (simp add: many_strings_of_length)
  have Tm: "card ?T = sum (%i. (2::nat) ^ i) {..r}"
    by (simp add: card_bitstrings_le_len)
  also have "... = sum (%i. (2::nat) ^ i) {..

```

Proof system locale

This locale states the information assumptions used in the argument. A bitstring p represents a proposed proof.

1. **Trace specification.** For any natural number k , if p proves that a particular positive n reaches 1, and

$$T^{(k)}(n) > 2,$$

then p contains an encoding of the parity vector

$$(n, T(n), \dots, T^{(k)}(n)) \pmod{2}.$$

2. **Universal instantiation.** If p proves the Collatz conjecture, then for every positive n the same certificate p proves the instance

$$\exists k. T^{(k)}(n) = 1.$$

The parity encoding is assumed injective.

Motivation for the trace-specification assumption

The trace-specification assumption is motivated by two structural properties of the Collatz map.

1. **Arbitrarily large excursions.** The affine representation

$$T^{(k)}(n) = \frac{3^s n + c}{2^k}$$

allows Collatz iterates to grow arbitrarily large relative to their starting values. For example, for every $k \geq 1$, starting at $n = 2^k - 1$ gives k consecutive increasing steps and reaches $3^k - 1$. The ratio

$$\frac{3^k - 1}{2^k - 1}$$

grows without bound as k increases.

This contrasts with the map

$$U(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ (n+1)/2 & \text{if } n \text{ is odd.} \end{cases}$$

For every $n > 1$, we have $1 \leq U(n) < n$, regardless of whether n is even or odd. Repeated application therefore reaches 1, without requiring an explicit parity trace.

2. **Injectivity of the affine representation.** In the Collatz affine representation, s counts the odd steps, while c depends on their positions. As shown earlier, for a fixed trace length k , the correspondence between parity vectors and their associated affine parameters (s, c) is injective.

Property 1 rules out the simple step-by-step descent argument used for U . The affine representation gives the exact formula for $T^{(k)}(n)$ associated with its parity vector. By property 2, its exact parameters and the parity vector determine one another at a fixed trace length. We therefore investigate proofs that explicitly store this parity information. The trace-specification assumption requires the proof certificate to contain the encoded parity trace

explicitly. The Isabelle development establishes the resulting limitation on such certificates.

```

locale Collatz_Trace_Barrier =
  fixes enc_parity :: "bool list  $\Rightarrow$  bitstring"
    and proves_reaches_one :: "bitstring  $\Rightarrow$  nat  $\Rightarrow$  bool"
    and is_collatz_proof :: "bitstring  $\Rightarrow$  bool"
    assumes enc_parity_injective: "inj enc_parity"

  assumes trace_specification:
    "[| proves_reaches_one p n;
      n > 0;
      Tpow k n > 2 |]
    ==> contains p
      (enc_parity (parity_vec n (Suc k)))"

  assumes collatz_proof_instances:
    "is_collatz_proof p ==> ALL n>0. proves_reaches_one p n"
begin

lemma incompressible_parity_encodings_exist:
  shows " $\exists s. \text{length } s = m \wedge \text{incompressible\_by } s \text{ enc\_parity}$ "
  using incompressible_strings_exist_for_enc[OF enc_parity_injective]
.

```

Interpretation of the main theorem

Suppose that p is a proof of the Collatz conjecture in the assumed system, and let $L = \text{length}(p)$. Choose a vector x of length $L + 1$ that is incompressible under the chosen parity encoding. The preceding realisability result supplies a positive n whose first $L + 1$ parity values equal x and whose parities at steps L and $L + 1$ are equal. Hence $T^{(L)}(n) > 2$, and if $T^{(k)}(n) = 1$, then $k > L$. Applying the trace-specification assumption at step L requires p to contain the encoded vector x . Its encoding has length at least $L + 1$, whereas every substring of p has length at most L . This is the required contradiction.

```

theorem no_finite_collatz_proof:
  assumes p_proof: "is_collatz_proof p"
  shows False
proof -
  define L where "L = length p"
  obtain x where
    x_len: "length x = Suc L" and
    x_incomp: "incompressible_by x enc_parity"
    using incompressible_parity_encodings_exist[of "Suc L"]
    by blast
  obtain n where
    n_pos: "n > 0" and

```



```

    pv_eq: "parity_vec n (Suc L) = x" and
    same_parity:
      "odd (Tpow (Suc L) n) = odd (Tpow L n)"
    using parity_vector_realizable_with_matching_next_parity[OF x_len]
    by blast
  have value_at_L_gt_two:
    "Tpow (length p) n > 2"
    using equal_successive_parity_imp_gt_two[OF n_pos same_parity]
    by (simp add: L_def)
  have instance_proof: "proves_reaches_one p n"
    using collatz_proof_instances[OF p_proof] n_pos
    by blast
  have pv_eq':
    "parity_vec n (Suc (length p)) = x"
    using pv_eq by (simp add: L_def)
  have contains_x: "contains p (enc_parity x)"
  proof -
    have trace_contained:
      "contains p
        (enc_parity (parity_vec n (Suc (length p))))"
      using trace_specification[
        OF instance_proof n_pos value_at_L_gt_two] .
      show ?thesis
        using trace_contained pv_eq' by simp
    qed
  have enc_large: "Suc L ≤ length (enc_parity x)"
    using x_incomp x_len
    by (simp add: incompressible_by_def compressible_def)
  have enc_fits: "length (enc_parity x) ≤ length p"
    using contains_len_bound[OF contains_x] .
  have "Suc L ≤ length p"
    using enc_large enc_fits by linarith
  then show False
    by (simp add: L_def)
qed

corollary no_collatz_proof_in_this_system:
  shows "¬ (∃ p. is_collatz_proof p)"
  using no_finite_collatz_proof
  by blast

end
end

```